



4.2 Functionals of a single function

<The simplest variation problem>

Class Ω		Type of Problem
x must be continuous	C	$J = \int g(x, t) dt$
x must be continuous with continuous first derivatives ($x \in D^1$)	D^1	$J = \int g(x, \dot{x}, t) dt$
x must be continuous with continuous first and second derivatives ($x \in D^2$)	D^2	$J = \int g(x, \dot{x}, \ddot{x}, t) dt$

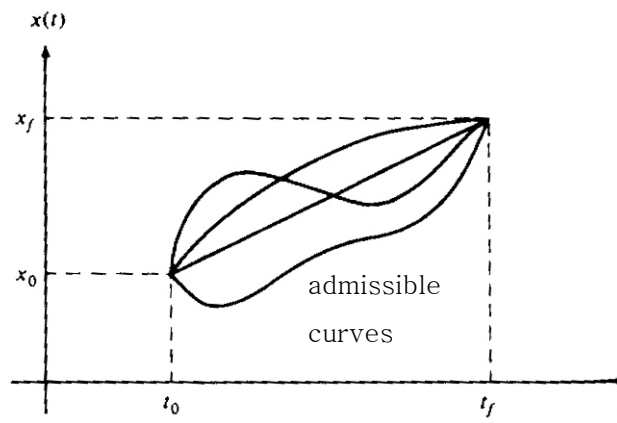


① $J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$

$x(t_0), x(t_f)$ are given, t_0 and t_f are fixed 

J is a functional of the function x .

Find the function x^* (extremal) for which the J has a relative extremum.



Curves in class Ω which also satisfy the end conditions are admissible.

The search begins by finding the curves that satisfy the fundamental theorem.

$$\Delta J(x^*, \dot{x}^*, t; \delta x, \delta \dot{x}) = J(x^* + \delta x, \dot{x}^* + \delta \dot{x}, t) - J(x^*, \dot{x}^*, t)$$

$$= \int_{t_0}^{t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt$$

Taylor series expansion about $x^, \dot{x}^*$*

$$= \int_{t_0}^{t_f} \left[g(x^*, \dot{x}^*, t) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} + H.O.T \right] dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt$$

$$H.O.T = \frac{1}{2} \left[\frac{\partial^2 g}{\partial x^2} \Big|_{x^*, \dot{x}^*} (\delta x)^2 + 2 \frac{\partial^2 g}{\partial x \partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta x \delta \dot{x} + \frac{\partial^2 g}{\partial \dot{x}^2} \Big|_{x^*, \dot{x}^*} (\delta \dot{x})^2 \right] + \theta [(\delta x)^2, (\delta \dot{x})^2]$$

We extract the first variation in ΔJ that are linear in δx and $\delta \dot{x}$

$$\Delta J \approx \delta J(x, \delta x) = \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right] dt$$

$$\text{integration by part } \int_{t_0}^{t_f} u v dt = uv \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} u v dt$$

$$= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x \right] dt$$

$$\delta x(t_0) = 0, \delta x(t_f) = 0 \text{ fixed (natural boundary condition)}$$

$$= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x(t) dt \underset{\text{fundamental theorem}}{=} 0$$

$\delta x(t)$ is arbitrary.

$$\therefore \frac{\partial g}{\partial x}(x^*, \dot{x}^*, t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t) \right] = 0$$

A necessary condition for x^* to be an extremal

Euler-Lagrange equation when $x(t_0), x(t_f)$ are given

Euler-Lagrange equation

Generally, nonlinear, ordinary, time-varying, second-order differential equation.

- usually cannot be solve analytically.
- numerical integration is used.
- the boundary condition are split.

$x(t_0), x(t_f)$ are known, instead of having $[x(t_0), \dot{x}(t_0)]$ or $[x(t_f), \dot{x}(t_f)]$

→ nonlinear, two-point boundary value problem (TPBVP)

This problem is difficult because of the combination of split boundary values and the nonlinearity of the differential equations.

$$\textcircled{2} \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), \ddot{x}(t), t) dt$$

t_0 and t_f are fixed

$x(t_0), x(t_f), \dot{x}(t_0), \dot{x}(t_f)$ are given

$$\delta J = \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \frac{\partial g}{\partial \ddot{x}} \delta \ddot{x} \right] dt$$

$$\text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right) dt = \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x dt$$

$$\text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{\partial g}{\partial \ddot{x}} \delta \ddot{x} \right) dt = \frac{\partial g}{\partial \ddot{x}} \delta \dot{x} \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} dt$$

$$= \frac{\partial g}{\partial \ddot{x}} \delta \dot{x} \Big|_{t_0}^{t_f} + \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} \right] dt$$

$$\text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} \right) dt = \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x dt$$

$$= - \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x + \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \right] dt$$

$$= \int_{t_0}^{t_f} \left[\frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \frac{\partial g}{\partial x} \right] \delta x dt$$

$$= 0$$

$$\therefore \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \frac{\partial g}{\partial x} = 0 \quad ; \text{Euler-Lagrange equation}$$

$$\textcircled{3} \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), \ddot{x}(t), \ddot{\ddot{x}}(t), t) dt$$

t_0 and t_f are fixed

$x, \dot{x}, \ddot{x}$ are given at both the end points.

$$\therefore - \frac{d^3}{dt^3} \left(\frac{\partial g}{\partial \ddot{\ddot{x}}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \frac{\partial g}{\partial x} = 0$$

Generalization

$$\therefore (-1)^n \frac{d^n}{dt^n} \left(\frac{\partial g}{\partial x} \right) + (-1)^{n-1} \frac{d^{n-1}}{dt^{n-1}} \left(\frac{\partial g}{\partial x} \right) + \dots + \frac{\partial g}{\partial x} = 0$$

Example (Low-Thrust Orbit Transfer (2D))

$$m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}$$



$\vec{r} = r\hat{e}_r$ (polar coordinate), $\vec{u}$: Thrust acceleration (control)

$$\phi = \phi(\vec{r}) \quad \text{gravitational potential, } \nabla\phi(\vec{r}) = \frac{\partial\phi(\vec{r})}{\partial\vec{r}} = \frac{\partial\phi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{e}_\theta$$

$$J = \frac{1}{2} \int_0^T (\vec{u}^T \vec{u}) dt$$

minimum – control – effort,

$\vec{r}(0), \vec{r}(T), \dot{\vec{r}}(0), \dot{\vec{r}}(T), T$ are specified

Find $\vec{u}$ to minimize J

Answer)

$$J(\vec{r}, \ddot{\vec{r}}) = \frac{1}{2} \int_0^T (\ddot{\vec{r}} + \nabla\phi)^T (\ddot{\vec{r}} + \nabla\phi) dt = \frac{1}{2} \int_{t_0}^{t_f} (\ddot{\vec{r}} + \nabla\phi)^2 dt$$

$$\Rightarrow J(\vec{r}, \ddot{\vec{r}}) = \int_0^T g(\vec{r}, \ddot{\vec{r}}) dt$$

Necessary Condition

$$\frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{\vec{r}}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{\vec{r}}} \right) + \frac{\partial g}{\partial \vec{r}} = 0$$



$$\frac{\partial g}{\partial \vec{r}} = \frac{\partial}{\partial \vec{r}} (\ddot{\vec{r}} + \nabla\phi)^2 = 2(\ddot{\vec{r}} + \nabla\phi) \frac{\partial}{\partial \vec{r}} (\nabla\phi(\vec{r})) = 2(\ddot{\vec{r}} + \nabla\phi) \nabla^2\phi(\vec{r})$$

$$\frac{\partial g}{\partial \dot{\vec{r}}} = 0$$

$$\frac{\partial g}{\partial \ddot{\vec{r}}} = \frac{\partial}{\partial \ddot{\vec{r}}} (\ddot{\vec{r}} + \nabla\phi)^2 = 2(\ddot{\vec{r}} + \nabla\phi) \frac{\partial}{\partial \ddot{\vec{r}}} (\ddot{\vec{r}}) = 2(\ddot{\vec{r}} + \nabla\phi)$$

$$\Rightarrow \frac{d^2}{dt^2}(\ddot{\vec{r}} + \nabla\phi) + \nabla^2\phi(\vec{r})(\ddot{\vec{r}} + \nabla\phi) = 0$$

$$\Rightarrow \ddot{\vec{u}} + \nabla^2\phi\vec{u} = 0, \quad E-L \text{ eq.}$$

$\vec{r}(0), \vec{r}(T), \dot{\vec{r}}(0), \dot{\vec{r}}(T), T$ are specified

$\Rightarrow$ Two-point Boundary-value Problem (TPBVP)

$$\Rightarrow \vec{u}^*(t)$$

$$m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}$$

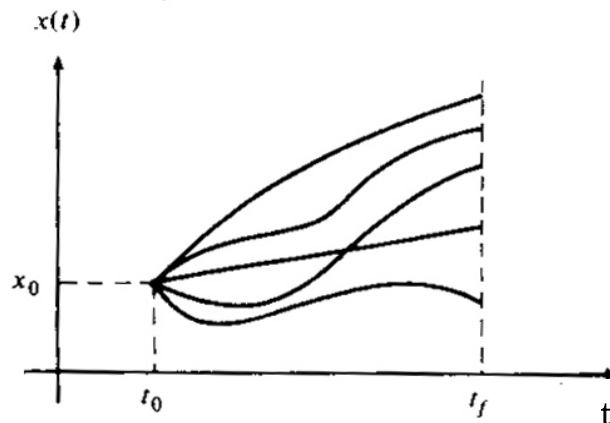
$$\Rightarrow \vec{r}^*(t)$$

< Final time specified, $x(t_f)$ free >

Find the necessary condition for a function to be an **extremal** for the functional

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

t_f is specified, $x(t_0)$ is given, $x(t_f)$ is free



$$\begin{aligned}
\delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt \\
&= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&\quad \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} = \frac{\partial g}{\partial \dot{x}} \delta x(t_f) - \frac{\partial g}{\partial \dot{x}} \delta x(t_0) = \frac{\partial g}{\partial \dot{x}} \delta x(t_f) \\
&= \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} \delta x(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= 0
\end{aligned}$$

For an extremal $x^*(t)$, we know $\delta J(\bar{x}^*, \delta \bar{x}) = 0$. Let us next show that

the integral of $\int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt$ must be zero on an extremal. Suppose that the curve $x^*(t)$ is an extremal for the free end point problem. Now consider a fixed end point problem with the same functional, the same initial and final times, and with specified end points that are the same as for the extremal $x^*(t)$ in the free end point problem. The curve $x^*(t)$ must be an extremal for this fixed end point problem; therefore, $x^*(t)$ must be a solution of the Euler equation, and the integral term must be zero on an extremal. In other words, an extremal for a free end point problem is also an extremal for the fixed end point problem with the same end points, and the same functional; thus, regardless of the boundary conditions, the Euler equation must be satisfied.

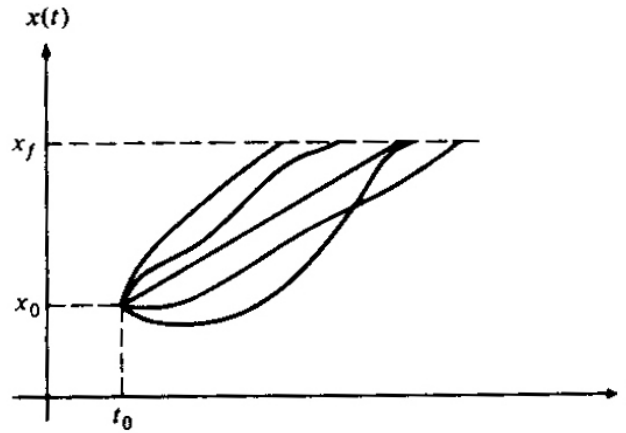
$$\begin{aligned}
\therefore \frac{\partial g}{\partial x}(x, \dot{x}, t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x, \dot{x}, t) \right] &= 0 && \text{Euler-Lagrange equation } 0 \leq t \leq t_f \\
\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} &= 0 && \text{at } t = t_f \quad \text{transversality condition} \\
&&& \text{(natural boundary condition)}
\end{aligned}$$

< Final time free, $x(t_f)$ specified >

Find the necessary condition for a function to be an extremal for the functional

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

t_0 is fixed, t_f is free, $x(t_0)$ and $x(t_f)$ are given



$$\begin{aligned} \Delta J &= \int_{t_0}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} g(x, \dot{x}, t) dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} [g(x, \dot{x}, t) - g(x^*, \dot{x}^*, t)] dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt \end{aligned}$$

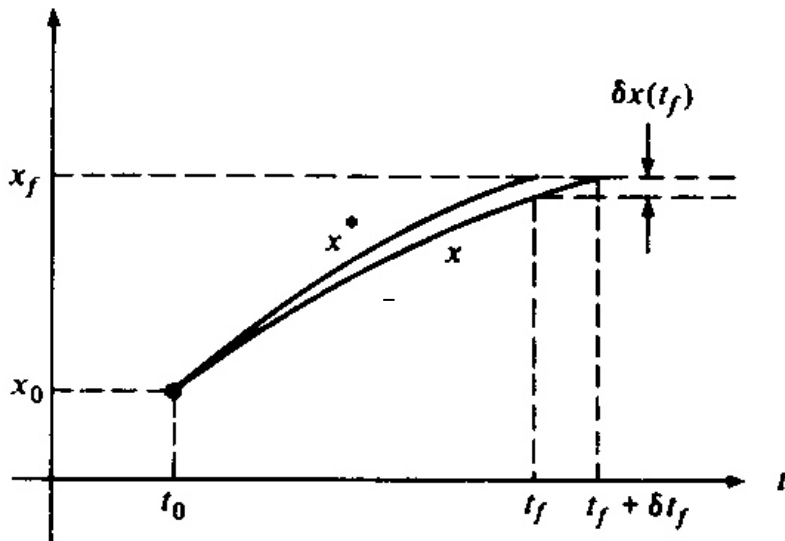
$$\int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt = g(x(t_f), \dot{x}(t_f), t_f) \delta t_f$$



$$\begin{aligned} &= \left[g(x^*(t_f), \dot{x}^*(t_f), t_f) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x(t_f) + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x}(t_f) + H.O.T \right] \delta t_f \\ &\approx g(x^*(t_f), \dot{x}^*(t_f), t_f) \delta t_f \end{aligned}$$



$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right] dt + g(x^*(t_f), \dot{x}^*(t_f), t_f) \delta t_f \\ &= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + g(x^*, \dot{x}^*, t_f) \delta t_f \end{aligned}$$



$$-\delta x(t_f) \approx \dot{x}(t_f) \delta t_f$$

$$\text{assume } \dot{x}(t_f) = \dot{x}^*(t_f)$$

$$\rightarrow -\delta x(t_f) \approx \dot{x}^*(t_f) \delta t_f$$

$$\rightarrow \delta x(t_f) \approx -\dot{x}^*(t_f) \delta t_f$$

$\delta x(t_f)$ depends on δt_f and is neither zero nor free.

The dependence of $\delta x(t_f)$ on δt_f must be linearly approximated because δJ consists of the first-order terms in the increment of ΔJ

$$\delta J = \left. \frac{\partial g}{\partial x} \right|_{x^*, \dot{x}^*} \delta x(t_f) - \left. \frac{\partial g}{\partial \dot{x}} \right|_{x^*, \dot{x}^*} \delta \dot{x}(t_f) + g(x^*, \dot{x}^*, t_f) \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right]_{x^*, \dot{x}^*} \delta x(t) dt$$

$$\delta x(t_f) \rightarrow -\dot{x}^*(t_f) \delta t_f$$

$$= \left[-\frac{\partial g}{\partial x} \dot{x}^*(t_f) + g \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right]_{x^*, \dot{x}^*} \delta x(t) dt, \quad \delta t_f, \delta x(t) \text{ are arbitrary}$$

$$= 0$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$

Euler-Lagrange equation

$$g(x^*, \dot{x}^*, t_f) - \frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t_f) \dot{x}^*(t_f) = 0 \quad \text{at } t = t_f \quad \text{Transversality condition}$$

Example $J = \int_0^{t_f} (\dot{x}^2 + x) dt$

$$x(0)=0, x(t_f)=1, \quad t_f \text{ is free}$$

$$g(x, \dot{x}, t_f) = (\dot{x}^2 + x)$$

① Euler-Lagrange equation

$$\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$



$$\rightarrow -2\ddot{x} + 1 = 0$$

$$\rightarrow \ddot{x} = \frac{1}{2}$$

$$\rightarrow x = At^2 + Bt + C$$

$$\rightarrow x(0) = 0 \rightarrow C = 0$$

$$\rightarrow x(t_f) = At_f^2 + Bt_f = 1$$

$$\rightarrow \ddot{x} = 2A \Rightarrow A = \frac{1}{4}$$

② Transversality condition

$$g - \frac{\partial g}{\partial \dot{x}} \dot{x} = 0 \quad \text{at } t = t_f$$

$$\dot{x}^2 + x - 2\dot{x}^2 = 0$$



$$x - \dot{x}^2 = 0$$

$$\dot{x} = 2At + B = \frac{1}{2}t + B$$

$$\dot{x}^2 = \frac{1}{4}t^2 + Bt + B^2$$

$$x = \dot{x}^2 \Big|_{t=t_f} \Rightarrow \frac{1}{4}t_f^2 + Bt_f = \frac{1}{4}t_f^2 + Bt_f + B^2$$

$$B^2 = 0 \rightarrow B = 0$$

$$\therefore x^*(t) = \frac{1}{4}t^2$$

$$x(t_f) = 1$$

$$\frac{1}{4}t_f^2 = 1$$

$$\rightarrow t_f = \pm 2$$

$$\therefore t_f = 2$$

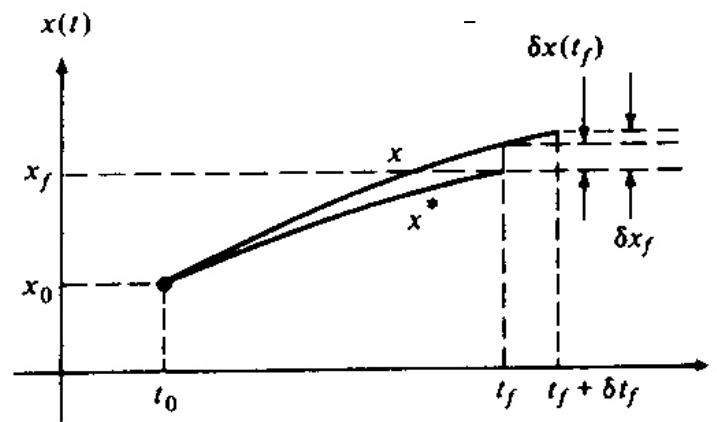
< Problems with both the final time t_f and $x(t_f)$ free >

Find a necessary condition that must be satisfied by an extremal for a function of the form.

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

t_0 and $x(t_0)$ are specified,

t_f and $x(t_f)$ are free



$\delta x(t_f)$; the difference at $t = t_f$.

the state variation at t_f .

δx_f ; the difference at the end point of the two curves.

the state variation when t_f and $x(t_f)$ are free.

In general, $\delta x(t_f) \neq \delta x_f$

$$\delta x_f \approx \delta x(t_f) + \dot{x}(t_f) \delta t_f$$

$$\dot{x}(t_f) \approx \dot{x}^*(t_f)$$

$$\delta x(t_f) \approx \delta x_f - \dot{x}^*(t_f) \delta t_f$$

$$\begin{aligned} \Delta J &= \int_{t_0}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} [g(x, \dot{x}, t) - g(x^*, \dot{x}^*, t)] dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt \end{aligned}$$

$$\begin{aligned}
\delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt + g(x^*(t_f), \dot{x}^*(t_f), t_f) \delta t_f \\
&= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + g(x^*, \dot{x}^*, t_f) \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
\frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} &= \frac{\partial g}{\partial \dot{x}} \delta x(t_f) - \frac{\partial g}{\partial \dot{x}} \delta x(t_0) \\
&= \frac{\partial g}{\partial \dot{x}} (\delta x_f - \dot{x}^* \delta t_f)
\end{aligned}$$

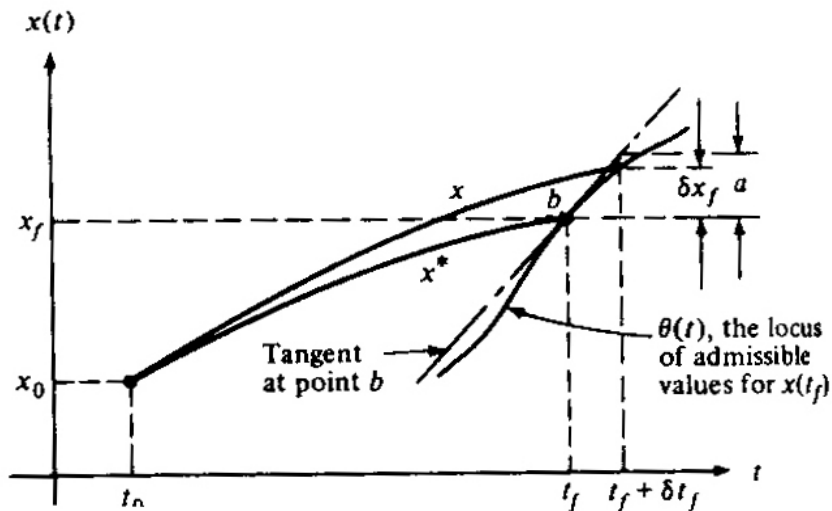
$$\begin{aligned}
&= \left[\frac{\partial g}{\partial \dot{x}} \delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \right] \\
&= 0
\end{aligned}$$

a) $x^*(t_f)$ and t_f^* are not related ($\delta x(t_f)$ and δx_f are arbitrary)

$$\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0 \quad \text{Euler-Lagrange equation}$$

$$\left. \begin{aligned} \frac{\partial g}{\partial \dot{x}} &= 0 \\ g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* &= 0 \end{aligned} \right\} \Rightarrow g(x^*, \dot{x}^*, t_f) = 0 \quad \text{at } t = t_f \quad \text{transversality condition}$$

b) $x^*(t_f)$ and t_f^* are related by $x^*(t_f) = \theta(t_f)$



$$\dot{\theta}(t_f) \approx \frac{\delta x_f}{\delta t_f}$$

$$\rightarrow \delta x_f \approx \dot{\theta}(t_f) \delta t_f$$

$$\begin{aligned} \delta J &= \frac{\partial g}{\partial \dot{x}} \dot{\theta}(t_f) \delta t_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^*(t_f) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= \left[g - \frac{\partial g}{\partial \dot{x}} (\dot{x}^* - \dot{\theta}(t_f)) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= 0 \end{aligned}$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0 \quad \text{Euler-Lagrange equation}$$

$$g + \frac{\partial g}{\partial \dot{x}} \left[\dot{\theta} - \dot{x}^* \right] \Big|_{x^*(t_f), \dot{x}^*(t_f), t_f} = 0 \quad \text{Transversality condition}$$

$$\Rightarrow g(x^*(t_f), \dot{x}^*(t_f), t_f) + \frac{\partial g}{\partial \dot{x}}(x^*(t_f), \dot{x}^*(t_f), t_f) [\dot{\theta}(t_f) - \dot{x}^*(t_f)] = 0$$

Example (see Problem 4-13 on p.181)

Determine an extremal for the functional

$$J(x) = \int_0^{t_f} \sqrt{1 + \dot{x}^2(t)} dt$$

$x(0) = 2$, $x(t)$ terminates on the curve $\theta(t) = -4t + 5$.
 t_f and $x(t_f)$ are free.

sol.)

$$g = \sqrt{1 + \dot{x}^2(t)}$$

$$x(t_f) = \theta(t_f) = -4t_f + 5$$

Euler-Lagrange equation

$$\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$

$$\rightarrow \frac{d}{dt} \left(\frac{\dot{x}}{(1 + \dot{x}^2)^{1/2}} \right) = 0$$

$$\rightarrow \frac{\ddot{x}}{(1 + \dot{x}^2)^{1/2}} - \frac{1}{2} \dot{x} \frac{2\dot{x}\ddot{x}}{(1 + \dot{x}^2)^{3/2}} = 0$$

$$\rightarrow \frac{\ddot{x}}{(1+\dot{x}^2)^{1/2}} \left[1 - \frac{\dot{x}^2}{(1+\dot{x}^2)} \right] = 0$$

$$\ddot{x}(t) = 0 \quad \left(\because 1 - \frac{\dot{x}^2}{(1+\dot{x}^2)} \neq 0 \right)$$

$$\Rightarrow x^*(t) = C_1 t + C_2, \quad (1 + \dot{x}^2 > 0) \quad (i)$$

from boundary condition

$$x(0) = 2 \rightarrow C_2 = 2$$

The transversality condition is

$$g + \frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) = 0 \quad \text{at } t = t_f$$

$$\rightarrow \frac{\dot{x}}{(1+\dot{x}^2)^{1/2}} [-4 - \dot{x}] + (1 + \dot{x}^2)^{1/2} = 0$$

$$\rightarrow -4\dot{x} - \dot{x}^2 + 1 + \dot{x}^2 = 0$$

$$\rightarrow -4\dot{x} + 1 = 0 \quad \text{at } t = t_f \quad (ii)$$

from (i) & (ii)

$$\dot{x}(t) = C_1 \rightarrow C_1 = \frac{1}{4}$$

The value of t_f is found from

$$x(t_f) = \theta(t_f)$$

$$\frac{1}{4}t_f + 2 = -4t_f + 5$$

$$\frac{17}{4}t_f = 3$$

$$t_f = \frac{12}{17}$$

$\therefore$ the solution

$$x^*(t) = \frac{1}{4}t + 2$$

$$t_f = \frac{12}{17}$$

Summary

1) Definition and phenomena

Functional, Extremal, Extremum

Relative minimum & maximum

$$\text{sign } \Delta J = \text{sign } \delta J$$

2) Fundamental theorem of calculus of variation

$$\delta J(\bar{x}^*, \delta x) = 0 \quad \text{for extremum}$$

$$3) \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt, \quad t_0, x(t_0) \text{ are given,}$$

The admissible curve $x(t)$ are smooth

$$\begin{aligned} \delta J(\bar{x}^*, \delta x) &= \frac{\partial g}{\partial \dot{x}} \delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= 0 \end{aligned}$$

We have found that regardless of the boundary conditions the Euler-Lagrange equation must be satisfied.

$$\therefore \frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t) \right] = 0$$

i) t_f and $x(t_f)$ are specified

$$\text{then, } \delta t_f = 0 \text{ and } \delta x_f = \delta x(t_f) = 0$$

ii) t_f is specified, $x(t_f)$ is free

$$\text{then, } \delta t_f = 0 \text{ and } \delta x_f = \delta x(t_f) \neq 0$$

$$\left. \frac{\partial g}{\partial \dot{x}} \right|_{t=t_f} = 0$$

iii) t_f is free, $x(t_f)$ is specified

then, $\delta t_f \neq 0$ and $\delta x_f = 0$

$$g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \Big|_{t=t_f} = 0$$

iv) t_f and $x(t_f)$ are free

then, $\delta t_f \neq 0, \delta x_f \neq 0$

a) $x^*(t_f)$ and t_f^* are not related

$$\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} = 0, \quad g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \Big|_{t=t_f} = 0 \quad \Rightarrow g \Big|_{t=t_f} = 0$$

b) $x^*(t_f) = \theta(t_f)$

$$\delta x_f \approx \dot{\theta} \delta t_f$$

$$\left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}^*) + g \right]_{t=t_f} = 0$$

So far, the functionals considered have contained only a single function (x) and its first derivative ($\dot{x}$). We can generalize our discussion to include functionals that may contain several independent functions and their first-derivatives.

⇒ The result are matrix version (Sec.4.3)

1) Regardless of the boundary conditions, the Euler-Lagrange equations are necessary conditions and must be satisfied.

$$\frac{\partial g}{\partial \vec{x}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right] = \vec{0}$$

2) The required boundary conditions are found from the following equation by making the appropriate substitutions for $\delta \vec{x}_f$ and δt_f .

$$\left[\frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right]_{t=t_f}^T \delta \vec{x}_f + \left[g(\vec{x}^*(t), \dot{\vec{x}}^*(t), t) - \left\{ \frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right\}^T \dot{\vec{x}}(t) \right]_{t=t_f} \delta t_f = 0$$

Table 4-1 DETERMINATION OF BOUNDARY-VALUE RELATIONSHIPS

<i>Problem description</i>	<i>Substitution</i>	<i>Boundary conditions</i>	<i>Remarks</i>
1. $\mathbf{x}(t_f)$, t_f both specified (Problem 1)	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f) = \mathbf{0}$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$	$2n$ equations to determine $2n$ constants of integration
2. $\mathbf{x}(t_f)$ free; t_f specified (Problem 2)	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = \mathbf{0}$	$2n$ equations to determine $2n$ constants of integration
3. t_f free; $\mathbf{x}(t_f)$ specified (Problem 3)	$\delta \mathbf{x}_f = \mathbf{0}$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)$ $-\left[\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)\right]^T \dot{\mathbf{x}}^*(t_f) = 0$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f
4. t_f , $\mathbf{x}(t_f)$ free and independent (Problem 4)	—	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = \mathbf{0}$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = 0$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f
5. t_f , $\mathbf{x}(t_f)$ free but related by $\mathbf{x}(t_f) = \boldsymbol{\theta}(t_f)$ (Problem 4)	$\delta \mathbf{x}_f = \frac{d\boldsymbol{\theta}}{dt}(t_f) \delta t_f^\dagger$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \boldsymbol{\theta}(t_f)$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)$ $+ \left[\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)\right]^T \left[\frac{d\boldsymbol{\theta}}{dt}(t_f) - \dot{\mathbf{x}}^*(t_f)\right] = 0^\dagger$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f

$^\dagger \frac{d\boldsymbol{\theta}}{dt}$ denotes the $n \times 1$ column vector $\left[\frac{d\theta_1}{dt} \quad \frac{d\theta_2}{dt} \quad \dots \quad \frac{d\theta_n}{dt}\right]^T$.